

Concept Steering in Gated DeltaNet: Low-Rank State Interventions and Contextual Decay

Anonymous Authors¹

Abstract

Linear-attention language models expose recurrent matrix-valued states that can be directly edited. We study concept steering in Qwen3.5-9B by injecting contrastive state differences into all Gated DeltaNet heads. Across three datasets of paired concepts, the updates have stable rank near two, while clamping their dominant singular component outperforms naive rank-one truncation. One-shot steering decays with context as contractive GDN updates erase the intervention, showing that effective control depends on both low-rank geometry and dynamical persistence.

1. Introduction

Activation steering provides a lightweight way to control a language model at inference time by modifying its internal representations (Turner et al., 2023; Zou et al., 2023; Panickssery et al., 2024). Most existing methods were developed for Transformer residual streams, where the intervention is a vector added at one or more layers. New linear-attention architectures expose a different intervention site: a recurrent state matrix that acts as a compressed associative memory. Editing this state is appealing for two reasons. First, it offers a mechanistically meaningful operation: steering changes the model’s stored key–value associations rather than adding an undifferentiated residual bias. Second, the recurrent update makes the lifetime of an intervention analytically tractable.

We investigate matrix-valued concept steering in Gated DeltaNet (Yang et al., 2025), using the text pathway of Qwen3.5-9B (Qwen Team, 2026). We estimate a per-head contrast matrix ΔS_h from paired concept and antagonist narratives, then add it to the GDN state. Our focal concepts span distinct output properties: optimistic versus pessimistic

content, Russian versus English generation, refusal versus compliance, and casual versus formal register. Unlike head-localized analyses, we intervene on every recurrent head in every GDN layer. This all-head intervention tests whether a distributed collection of low-dimensional state edits can induce a coherent behavioral change.

The experiments address four questions:

1. What is the effective rank of a contrastive GDN state update, and which singular components carry steerable information?
2. How quickly is a one-shot state intervention forgotten as unrelated context is processed?
3. Can several state-space concepts be composed without destructive interference?
4. How does steering strength trade off against general instruction-following quality?

Our main findings are:

- Contrastive state matrices are extremely low effective rank: across 100 concepts their mean stable rank is approximately two, while roughly six to eight singular directions explain 90% of squared spectral mass.
- Increasing the number of contrastive pairs reduces the estimated rank after a small-sample transient, consistent with averaging away example-specific components.
- Plain rank-one truncation can discard behaviorally useful components. In contrast, repeatedly clamping the dominant singular association prevents recurrent erasure and yields stronger steering.
- A one-shot perturbation decays with context according to the GDN transition dynamics. The latent magnitude contracts smoothly, while output behavior exhibits concept-dependent thresholds.

These results extend representation engineering from vector activations to matrix-valued recurrent memory. They also

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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expose a central distinction: *representation rank* describes how compactly a concept is encoded, whereas *dynamical persistence* determines whether that concept survives generation.

2. Background and Related Work

Gated DeltaNet. Delta-rule linear attention treats its recurrent matrix as an associative memory and performs targeted overwrites rather than purely additive updates (Yang et al., 2024). Gated DeltaNet combines this delta rule with data-dependent forgetting (Yang et al., 2025). For one head and token representation x_t , we use the following notation:

$$\begin{aligned} q_t &= W_q x_t, & v_t &= W_v x_t, \\ k_t &= \frac{W_k x_t}{\|W_k x_t\|_2}, \end{aligned} \quad (1)$$

$$\beta_t = \sigma(w_\beta^\top x_t), \quad \alpha_t = \exp[-\text{softplus}(w_\alpha^\top x_t)]. \quad (2)$$

Both gates lie in $(0, 1)$. Defining

$$A_t = \alpha_t I - \beta_t k_t k_t^\top, \quad (3)$$

the state and head output are

$$S_t = S_{t-1} A_t + \beta_t v_t k_t^\top, \quad (4)$$

$$o_t = S_t q_t, \quad (5)$$

$$y_t = W_o(\text{norm}(o_t) \odot \text{silu}(g_t)), \quad g_t = W_g x_t. \quad (6)$$

Thus $S_t \in \mathbb{R}^{d_v \times d_k}$ stores value directions in its rows and key-addressable directions in its columns. Qwen3.5-9B contains 32 layers in a 3:1 hybrid layout: 24 GDN layers and eight gated-attention layers. Its GDN head dimension is 128, with 32 value heads and 16 query/key heads (Qwen Team, 2026).

Representation steering. Activation Addition and representation engineering showed that high-level behaviors can be modified with linear directions (Turner et al., 2023; Zou et al., 2023). Contrastive Activation Addition estimates such a direction by averaging activation differences over positive and negative examples (Panickssery et al., 2024). Low-dimensional behavioral structure is especially visible for refusal, which can be mediated by a single residual-stream direction in several Transformer families (Arditi et al., 2024). We adopt the same contrastive principle but apply it to a matrix-valued recurrent memory. Consequently, both left and right singular subspaces—what is written and how it is addressed—can matter.

3. Matrix-Valued Concept Steering

3.1. Contrastive state construction

Let $S_{h,i}^+$ and $S_{h,i}^-$ be the terminal state of recurrent head h after processing the two members of matched pair i . The

positive member expresses the target concept and the negative member its antagonist. For N pairs, the steering matrix is

$$\Delta S_h = \frac{1}{N} \sum_{i=1}^N S_{h,i}^+ - \frac{1}{N} \sum_{i=1}^N S_{h,i}^-. \quad (7)$$

At intervention time τ , we update every GDN state head as

$$S_{h,\tau} \leftarrow S_{h,\tau} + c \Delta S_h, \quad (8)$$

where $c \in \mathbb{R}$ controls strength and sign. The main context-length experiment sets $\tau = 0$, before processing any prefix, so that the initial zero state is replaced by the concept memory. We also implement post-prefill and generation-time refresh schedules; Section B formalizes these variants.

3.2. Rank diagnostics

Write the singular value decomposition

$$\begin{aligned} \Delta S_h &= U_h \Sigma_h W_h^\top \\ &= \sum_{j=1}^{128} \sigma_{h,j} u_{h,j} u_{h,j}^\top, \quad \sigma_{h,1} \geq \dots \geq \sigma_{h,128}. \end{aligned} \quad (9)$$

We measure effective dimension using stable rank and the number of singular components required to preserve 90% of squared spectral mass:

$$\text{srank}(\Delta S_h) = \frac{\|\Delta S_h\|_F^2}{\|\Delta S_h\|_2^2}, \quad (10)$$

$$\text{rank}_{90}(\Delta S_h) = \min \left\{ k : \frac{\sum_{j=1}^k \sigma_{h,j}^2}{\sum_{j=1}^{128} \sigma_{h,j}^2} \geq 0.9 \right\}. \quad (11)$$

Stable rank is continuous and robust to small nonzero singular values, while rank_{90} describes the size of a useful truncated approximation.

3.3. Full, rank-one, and clamped interventions

We compare three state interventions:

$$\begin{aligned} \text{full:} & \quad c \Delta S_h, \\ \text{rank-one:} & \quad c \sigma_{h,1} u_{h,1} u_{h,1}^\top, \\ \text{clamped rank-one:} & \quad S_t \leftarrow S_t + u (c \sigma w^\top - u^\top S_t). \end{aligned} \quad (12)$$

For readability, the head index and leading-component subscript are omitted in the final line. Because $\|u\|_2 = 1$, Equation (12) enforces $u^\top S_t = c \sigma w^\top$ exactly after each correction. It therefore preserves one learned value-key association while leaving the component of S_t orthogonal to u unchanged. This is not merely a low-rank approximation: it changes the dynamics by restoring the selected association after recurrent forgetting.

3.4. Predicted decay of a one-shot intervention

Consider a steered and unsteered copy of one head following the same sequence, and hold their token-dependent $q_t, k_t, v_t, \alpha_t, \beta_t$ fixed. The additive write in Equation (4) cancels between the copies, giving

$$\Delta S_t = \Delta S_{t-1} A_t = \Delta S_0 \prod_{\ell=1}^t A_\ell. \quad (13)$$

Since k_t is unit norm, A_t has eigenvalue $\alpha_t - \beta_t$ along k_t and α_t on its orthogonal complement. Hence

$$\|A_t\|_2 = \max\{\alpha_t, |\alpha_t - \beta_t|\} < 1, \quad (14)$$

and

$$\|\Delta S_t\|_2 \leq \|\Delta S_0\|_2 \prod_{\ell=1}^t \|A_\ell\|_2. \quad (15)$$

The perturbation is therefore contractive in this frozen-trajectory analysis. Moreover, right multiplication cannot increase rank. For a rank-one injection, the left singular direction remains fixed while the right factor is continually transformed and contracted. In the full network the two trajectories can diverge because steering changes later activations, so Equation (13) is a local mechanistic model rather than an exact end-to-end identity. It nevertheless predicts the decay observed below.

4. Experimental Setup

Model and intervention scope. All experiments use Qwen3.5-9B and text-only prompts. We intervene on all recurrent heads in all 24 GDN layers; gated-attention layers and model parameters are unchanged. The reported results therefore characterize a distributed state intervention rather than a head-local edit.

Contrastive data. We use the Feature Stories dataset (Kornikov, 2026), which contains matched narratives with a shared setup and opposite behavioral, affective, or epistemic poles. The broad rank audit samples 100 concept-antagonist pairs. Detailed experiments use four focal contrasts: optimism/pessimism, Russian/English, refusal/non-refusal, and casual/formal language. The largest extraction sets contain on the order of two thousand paired examples per focal task. Because the corpus is synthetic and spans multiple languages and genres, conclusions about rank should be interpreted as properties of both the model and this contrastive distribution.

Behavioral metrics. Russian steering is scored by the fraction of output detected as Russian; casual register and refusal are scored by task-specific detectors; optimism is rated on a 1–5 scale by a fixed LLM judge. For abstract concepts

we use the same judge prompt and decoding settings across steering conditions. We measure general capability with IFEval, a benchmark of objectively verifiable instruction-following constraints (Zhou et al., 2023). Steering strength c is swept separately for each method and concept; the relevant object is the Pareto trade-off between concept score and IFEval, not a universal raw coefficient.

Context-length protocol. We prepend unrelated Wikipedia passages with lengths 0, 128, . . . , 8192 tokens to the evaluation prompt. The principal GDN condition injects ΔS_0 once into the zero initial state. We compare against the unsteered model and a conventional residual-stream steering baseline. Abstract traits are judged by an LLM, while language is measured directly. The refusal contrast is omitted from this experiment. We additionally log the leading singular magnitude of the propagated correction around the prefill-to-generation boundary.

5. Results

5.1. Concept differences are low effective rank

Figure 1 shows the per-head rank distributions for the four focal contrasts and the mean rank for 100 sampled concepts. Most per-head stable ranks lie between one and three. Across concepts, the mean stable rank is tightly concentrated around two, despite each state matrix having dimension 128×128 . The corresponding mean rank₉₀ is roughly 5.5–8.2. Thus the matrices are not algebraically rank one, but their energy is concentrated in a small subspace.

The low-rank structure is not simply a consequence of using too few examples. The additional convergence curves in Section C and Figure 8 vary the number of contrastive pairs. Some concepts show a small initial rise from one to a handful of pairs, when additional semantic variation first enters the estimate. Beyond that transient, both stable rank and rank₉₀ decrease and plateau with sample size. At the largest sample sizes, stable rank is approximately 1.55–1.9 and rank₉₀ approximately 3–5.5 across the four contrasts. This trend is consistent with a shared low-rank concept signal plus idiosyncratic example-level components that cancel under averaging, and motivates using the largest available extraction sets.

5.2. Low rank does not make truncation lossless

Figure 2 compares the behavioral effects of the full matrix, its leading singular component, and the clamped leading component. Plain rank-one steering loses substantial effect for Russian (90.0% to 30.0%), casual register (40.0% to 33.3%), and refusal (92.0% to 81.0%). Optimism is comparatively insensitive (4.3 versus 4.2 on the five-point judge scale). Therefore, a stable rank near two does not imply that

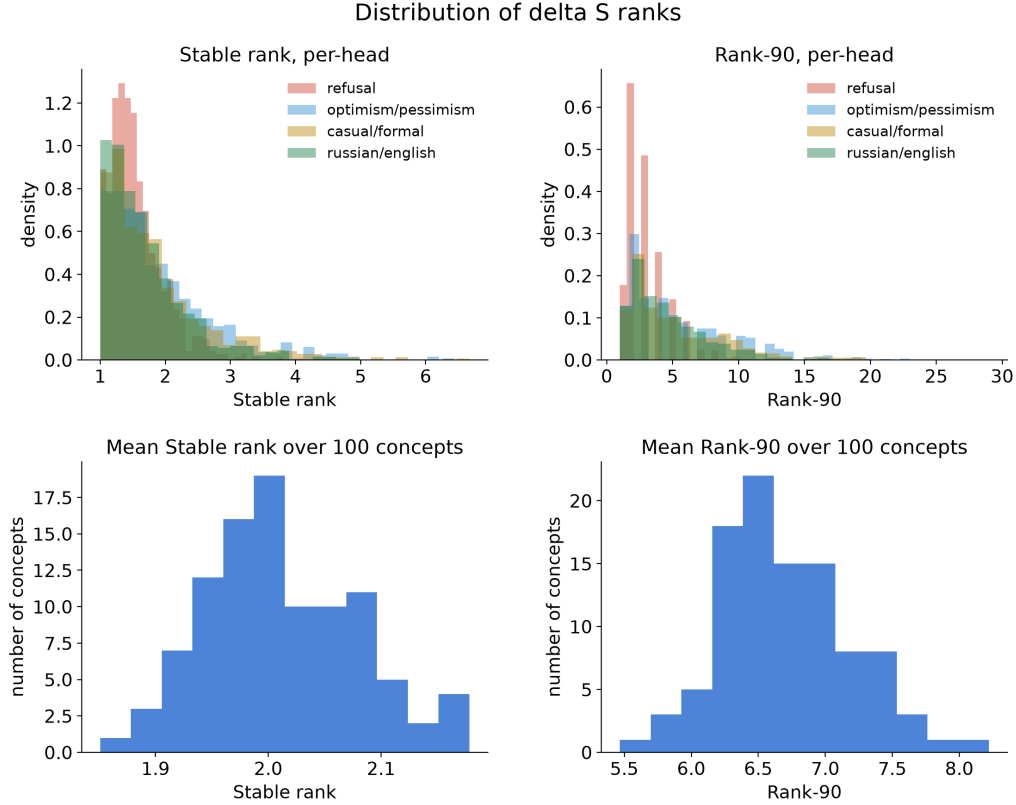


Figure 1. Effective rank of contrastive state matrices. Top: per-head distributions for the four focal concepts. Bottom: distributions over the mean per-head rank of 100 concept pairs. Stable rank concentrates near two; only a small number of the 128 singular directions is required to explain 90% of squared spectral mass.

the first singular component alone contains all behaviorally relevant information.

Clamping changes the result. Rank-one clamp reaches 96.7% Russian, 96.7% casual, a 4.3 optimism score, and 96.0% refusal. It is strongest or tied for strongest on every task. The gap between ordinary and clamped rank-one steering indicates that the dominant component is useful but is rapidly modified by the recurrence. Reinstating that component can matter more than retaining all weaker singular directions only at the initial intervention.

5.3. One-shot steering decays with unrelated context

Figure 3 reports behavioral steering as prefix length grows. The one-shot GDN intervention weakens for all three concepts, but the output transition is not universal. Russian remains near saturation through a few hundred tokens, then falls sharply after 1k tokens and reaches zero at 8k. Optimism decays more gradually, remaining above the unsteered baseline at 8k. Casual-register steering is the most fragile: it is close to baseline by 128 tokens. Residual steering applied near generation remains comparatively stable because it is not required to survive the entire prefill recurrence.

Figure 4 separates the latent contraction from the nonlinear output metric for Russian/English. The leading singular magnitude decreases systematically with the amount of prefill, so longer contexts arrive at the first generated token with a smaller remaining correction, matching the prediction of Equations (13) and (15). The broadly smooth contraction in state space can produce a delayed cliff in output space when the remaining signal crosses the model’s threshold for expressing the concept. The same pattern is reproduced for optimism/pessimism in Section E and Figure 10.

We also observe a sharp, reproducible drop at the boundary where generation begins. This is consistent with a change in the transition matrices A_t when the input distribution shifts from externally supplied prefix tokens to self-generated tokens, but the current evidence does not establish that cause. A kernel or cache-handling artifact could produce a similar signature. Disambiguating these explanations requires logging α_t , β_t , and key alignment under matched teacher-forced and decode-mode executions (see Section F and Figure 11).

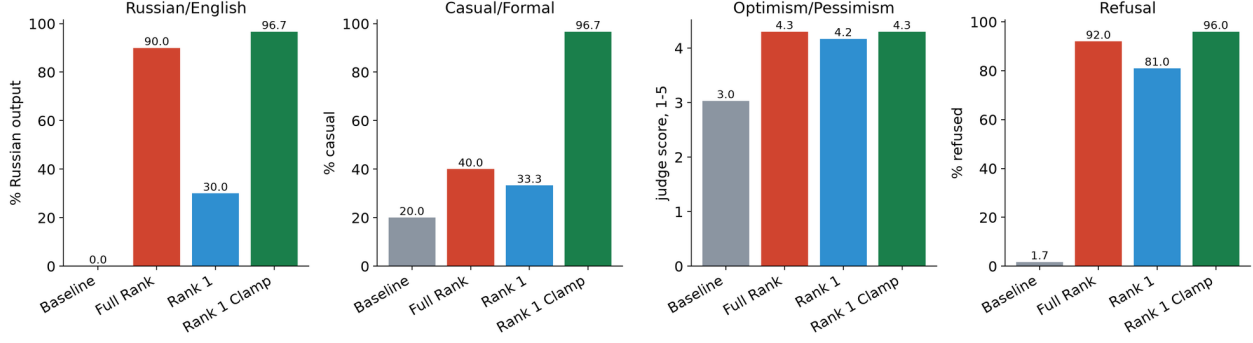


Figure 2. Concept strength under different state interventions. Metrics are concept-specific: Russian-output rate, casual-register rate, an optimism judge score from 1–5, and refusal rate. Rank-one truncation is weaker than the full matrix on three tasks, while tokenwise rank-one clamping is best or tied across all four. Because metric scales and steering coefficients differ by concept, comparisons should be made within each panel.

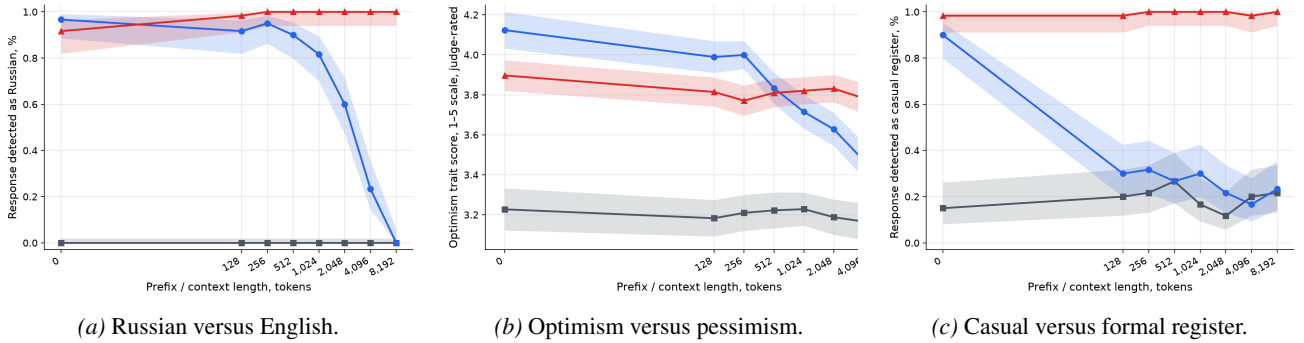


Figure 3. Behavioral steering versus unrelated prefix length. Gray denotes the unsteered baseline, blue a one-shot GDN state intervention at token zero, and red a residual-stream baseline applied near generation. Shaded bands show the reported dispersion across prompts. Latent contraction manifests as concept-dependent behavioral thresholds rather than a single universal decay curve.

5.4. Instruction-following quality and steering strength

We sweep strength for each steering family and evaluate Russian-language rate alongside IFEval prompt-strict accuracy. Figure 5 shows the resulting trade-offs. Clamped rank-one GDN produces the strongest GDN control: at strength 2.4, it reaches a Russian rate of 0.88 with IFEval accuracy of 0.56. Full-rank GDN reaches a lower peak Russian rate (0.74) with IFEval near 0.63, while ordinary rank-one GDN retains higher IFEval at the cost of a weaker maximum language shift. Residual steering reaches 0.89 Russian rate with IFEval accuracy of 0.67 at its selected high-control point. Thus, rank-one clamp is the strongest tested recurrent-state intervention, whereas residual steering retains more instruction-following performance at similarly high language control.

These results argue against choosing c solely to maximize the target score. The preferred operating point is the knee of the concept-IFEval frontier, and it is concept dependent. Raw coefficients are also not comparable across concepts because $\|\Delta S_h\|$ varies by head and dataset. A complete deployment should therefore report both c and the induced

state norm relative to the unsteered state.

5.5. Composing multiple concepts

For M requested concepts, we retain rank r separately for each head and concept, then form either a raw or normalized sum:

$$\Delta S_h^{(m,r)} = \sum_{j=1}^r \sigma_{h,j}^{(m)} u_{h,j}^{(m)} (w_{h,j}^{(m)})^\top, \quad r \in \{1, 4\}. \quad (16)$$

$$\Delta S_h^{\text{raw}} = \sum_{m=1}^M c_m \Delta S_h^{(m,r)}, \quad (17)$$

$$\Delta S_h^{\text{RSS}} = \Delta S_h^{\text{raw}} \frac{\sqrt{\sum_{m=1}^M \|c_m \Delta S_h^{(m,r)}\|_F^2}}{\|\Delta S_h^{\text{raw}}\|_F + \epsilon}. \quad (18)$$

The root-sum-square (RSS) factor preserves the aggregate norm implied by the individual edits while accounting for cancellation or reinforcement in their raw sum. Section D and Figure 9 provides additional geometric and condition-level examples of this normalization.

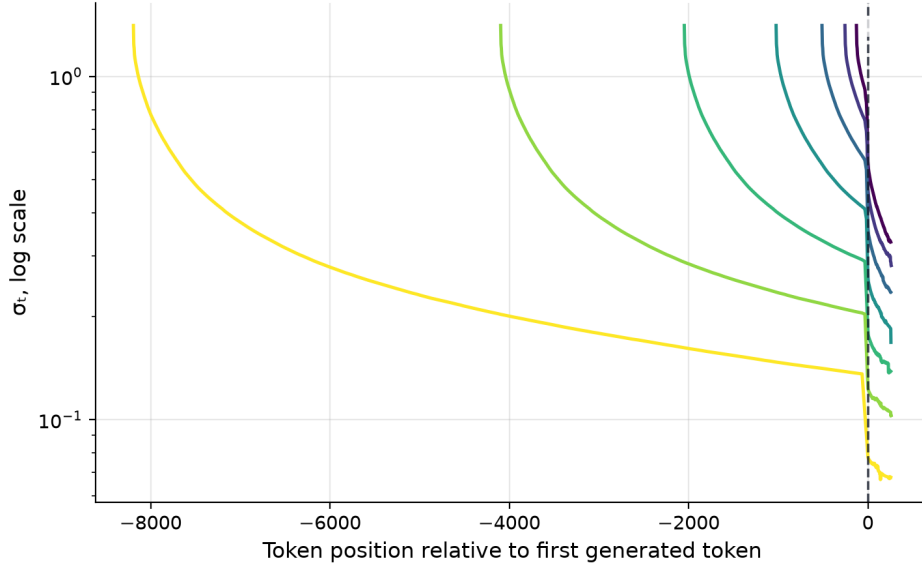


Figure 4. Effective magnitude of the propagated Russian/English state correction across prefill and generation (log vertical scale). Curves range from 128-token prefixes (purple) to 8192-token prefixes (yellow); token zero is the first generated token. Longer prefixes contract the intervention more strongly, providing direct state-space evidence for contextual decay.

Four-concept comparison. We compose four behavioral targets and evaluate the minimum expected score over the targets, which penalizes methods that improve only the easiest feature. In the complete sweep, rank-four GDN with RSS normalization achieves the highest minimum score (3.089) and four-feature success rate (12.5%), compared with 2.900 and 9.4% for RSS-normalized residual steering and 2.897 and 6.2% without steering. In the matched rank-one comparison (Figure 6), GDN improves the worst-target score over residual steering under raw composition (+0.091, 95% CI [+0.016, +0.172]) and RSS normalization (+0.093, 95% CI [+0.008, +0.178]).

Cross-feature interactions. Composition is not uniformly additive. In the factorial Russian, optimism, casualness, and refusal experiment, all diagonal effects are positive, but off-diagonal effects are selective (Figure 7). Russian steering reduces optimism and casualness success by 0.17 and 0.24; optimism raises casualness by 0.15 while reducing refusal by 0.16; and refusal reduces optimism and casualness by 0.15 and 0.17. With all four directions active, Russian, optimism, casualness, and refusal are detectable in 89.9%, 66.4%, 61.3%, and 28.6% of responses, respectively. Rank and normalization should therefore be treated as separate design choices, and joint steering should be evaluated per feature rather than only by a scalar aggregate.

6. Discussion

Compact representation versus persistent control. The rank experiments and clamp experiments measure different

properties. A concept can occupy a compact subspace yet be easy for the recurrence to erase. Conversely, weaker singular directions may be useful because they address different keys, even when they explain little Frobenius energy. Interpreting singular value as causal importance is therefore unsafe without intervention.

Distributed implementation. Low rank within a head does not imply localization to a single head or layer. The all-head intervention is a distributed collection of small, low-dimensional state changes. A useful next step is structured ablation over layers and heads to identify whether that distribution is redundant, synergistic, or localized to a smaller causal subset.

Safety. Refusal steering demonstrates that recurrent-state editing can modify aligned behavior. As with residual refusal directions (Arditi et al., 2024), this creates both diagnostic value and misuse risk. State intervention should be evaluated not only for target control and IFEval but also for unintended safety transfer, over-refusal, and interactions with adversarial prompts.

Limitations. The study uses one model family, synthetic contrastive narratives, four detailed concepts, and automated behavioral evaluators. The context study extends only to 8k tokens even though the model supports substantially longer contexts. Exact decay constants are not fitted, and the generation-boundary drop remains observational. The frozen-trajectory derivation ignores feedback from the intervention into later gates and keys. Finally, the composition

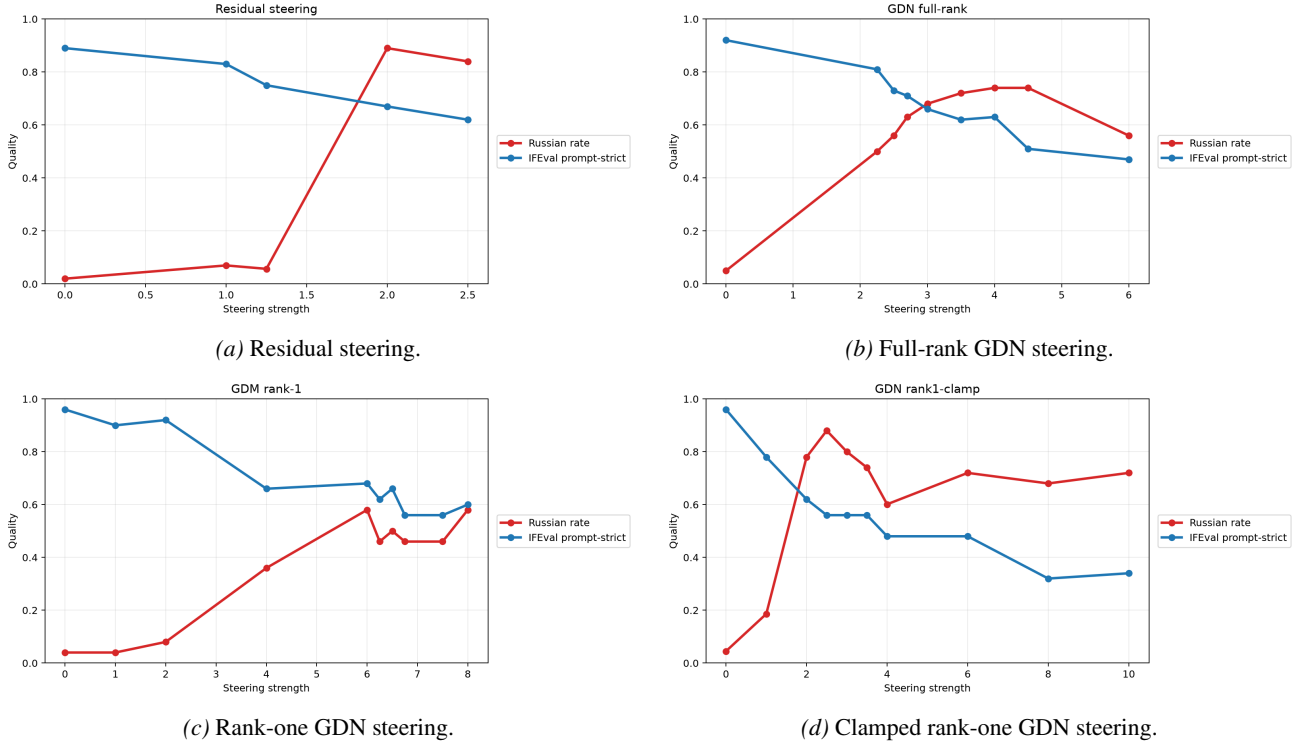


Figure 5. Control–quality trade-off as steering strength varies. Red is the Russian-language rate; blue is IFEval prompt-strict accuracy. Raw strength scales are method-specific, so comparisons should focus on the attained control–quality pairs rather than horizontal positions.

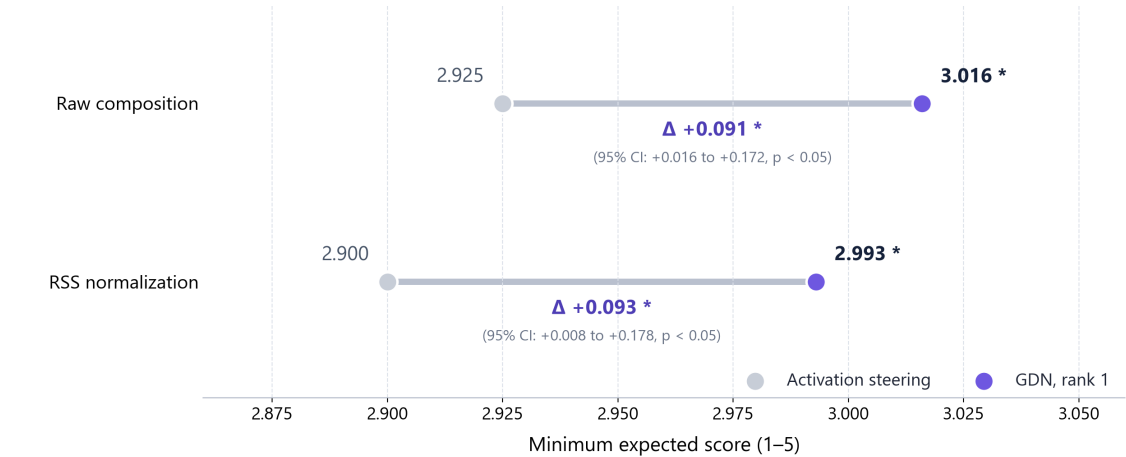
experiments cover only two four-concept sets and do not establish robust scaling to larger or semantically dissimilar sets.

7. Conclusion

Contrastive concept information in Qwen3.5-9B Gated DeltaNet states is strikingly low rank, but steering quality depends on dynamics as well as geometry. Plain rank-one truncation can weaken control, whereas clamping the dominant association counteracts recurrent erasure and yields the best observed GDN control–quality trade-off. One-shot interventions decay as context accumulates, in agreement with the contractive state recurrence, and different concepts cease to affect generation at different effective magnitudes. Multi-concept experiments show that GDN composition can improve the weakest requested feature relative to residual steering, although feature-specific interference remains. Matrix-valued state steering is therefore a promising mechanistic tool, provided that persistence, composition, and capability preservation are treated as first-class design objectives.

Matched comparison with ordinary activation steering

Minimum expected score across the same four target features



* Statistically significant gain ($p < 0.05$)

Figure 6. Matched four-concept comparison with conventional residual-stream steering. Each point is the minimum expected feature score (1–5) over the same requested targets; stars denote the reported $p < 0.05$ comparison. Rank-one GDN improves this worst-target metric under both raw and RSS composition.

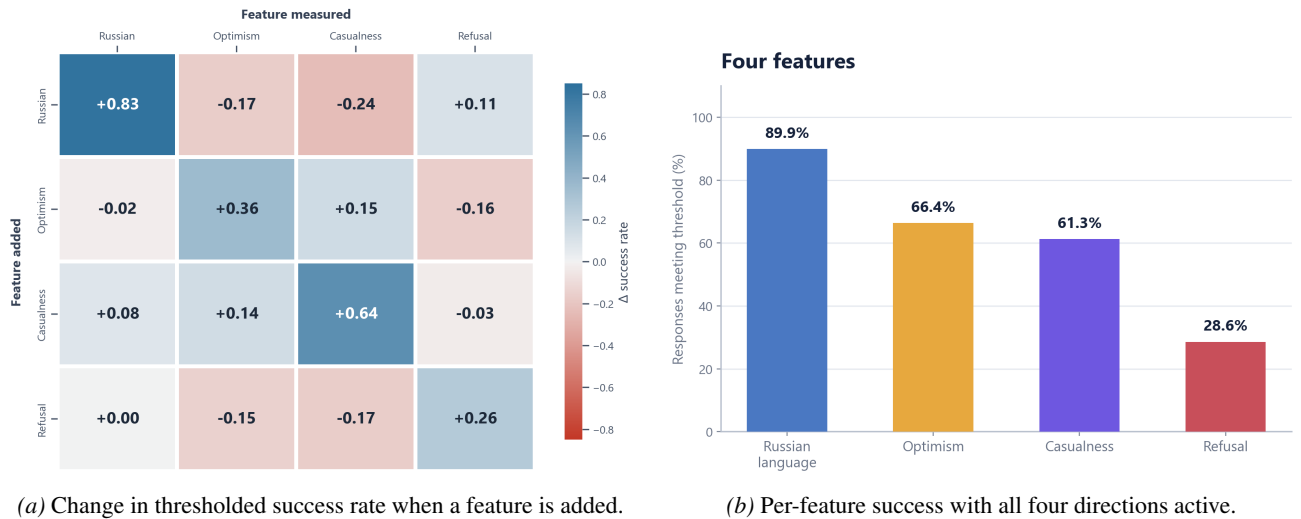


Figure 7. Multi-concept steering has strong on-target effects but selective cross-feature interference. The matrix distinguishes the added feature (rows) from the feature measured (columns).

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A. Derivation of Perturbation Propagation

Let S'_t denote the steered state and S_t the reference state. Under the fixed-trajectory assumption, both receive the same transition A_t and write $B_t = \beta_t v_t k_t^\top$:

$$S'_t = S'_{t-1} A_t + B_t, \quad (19)$$

$$S_t = S_{t-1} A_t + B_t. \quad (20)$$

Subtracting gives $\Delta S_t = \Delta S_{t-1} A_t$ and induction yields Equation (13). Since $k_t k_t^\top$ is an orthogonal rank-one projector, A_t is symmetric with spectrum $\{\alpha_t - \beta_t, \alpha_t, \dots, \alpha_t\}$. This establishes Equation (14). Submultiplicativity of the spectral norm gives Equation (15).

For $\Delta S_0 = c \sigma u w^\top$, define

$$r_t^\top = w^\top \prod_{\ell=1}^t A_\ell. \quad (21)$$

Then $\Delta S_t = c \sigma u r_t^\top$ remains rank at most one and

$$\|\Delta S_t\|_2 = |c| \sigma \|r_t\|_2. \quad (22)$$

This form clarifies the clamp: Equation (12) resets the row of the state addressed by u to the original right factor $c \sigma w^\top$, rather than allowing r_t to contract and rotate indefinitely.

B. Intervention Schedules

We distinguish three schedules:

Initial-state injection. Apply Equation (8) to S_0 before prefill. This is the main context-length condition and directly tests memory persistence.

Post-prefill injection. Apply the same update once after the full prompt and immediately before the first generated token. This removes prefill decay while retaining state-space intervention.

Generation refresh. Reapply the update, or the clamp in Equation (12), after each generated-token update. This tests persistent control and incurs an intervention at every decode step.

The schedules answer different questions and should not be conflated: initial-state injection measures whether GDN memory retains a concept; post-prefill injection measures immediate causal efficacy; and refresh measures achievable persistent control.

C. Additional Rank Diagnostics

The following convergence curves support the main rank-distribution result by showing that the low-rank estimates stabilize as the number of matched contrastive pairs grows.

D. Additional Composition Results

These diagnostics visualize RSS normalization and selected partial compositions. They complement, rather than repeat, the main four-concept method comparison and cross-feature interference results.

E. Additional State-Magnitude Diagnostics

The following plot repeats the main effective-magnitude analysis for the optimism/pessimism contrast. Its similar trajectory shows that the decay is not specific to the Russian/English intervention.

F. Generation-Boundary Diagnostics

Figure 11 enlarges the region around the first generated token. The contraction is visible across all tested prefix lengths and both concept pairs. A targeted follow-up should (i) replay identical tokens through prefill and decode kernels, (ii) teacher-force the continuation so the token sequence is controlled, (iii) decompose the change into α_t , β_t , and k_t alignment, and (iv) verify that state serialization and precision are identical across execution modes.

G. Cosine Diagnostics

We also track alignment between the propagated rank-one factor and the current query. These diagnostics are not used as primary evidence because their relationship to output behavior is concept dependent. Russian/English shows increased positive alignment during generation, especially for shorter contexts, whereas optimism/pessimism remains close to zero and changes sign around the boundary. The mismatch reinforces that correction magnitude and query alignment are distinct factors in the readout $\Delta S_t q_t$.

ΔS rank convergence vs. number of pairs

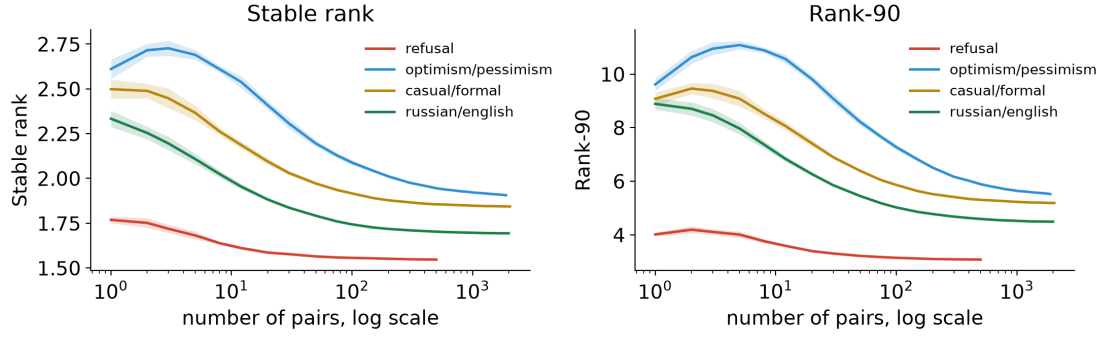


Figure 8. Rank convergence as the number of matched pairs increases. After a small-sample transient, larger contrast sets produce lower and more stable rank estimates, consistent with cancellation of example-specific noise. Shaded regions show the reported dispersion across heads.

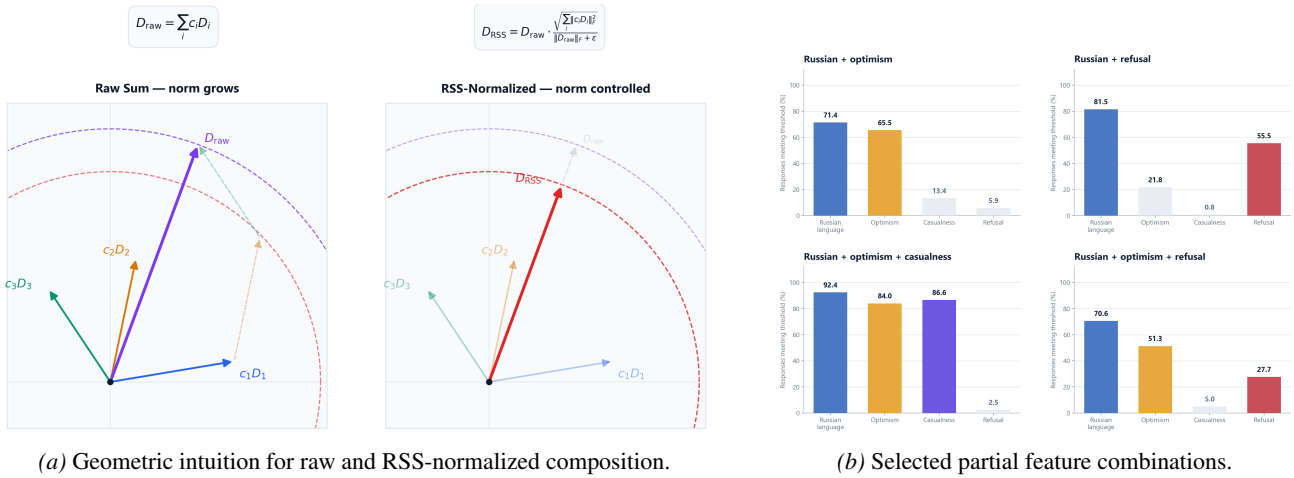


Figure 9. Composition mechanics and condition-level examples. RSS rescales the update after vector addition; the bar charts show both target retention and selective interference before all four directions are combined.

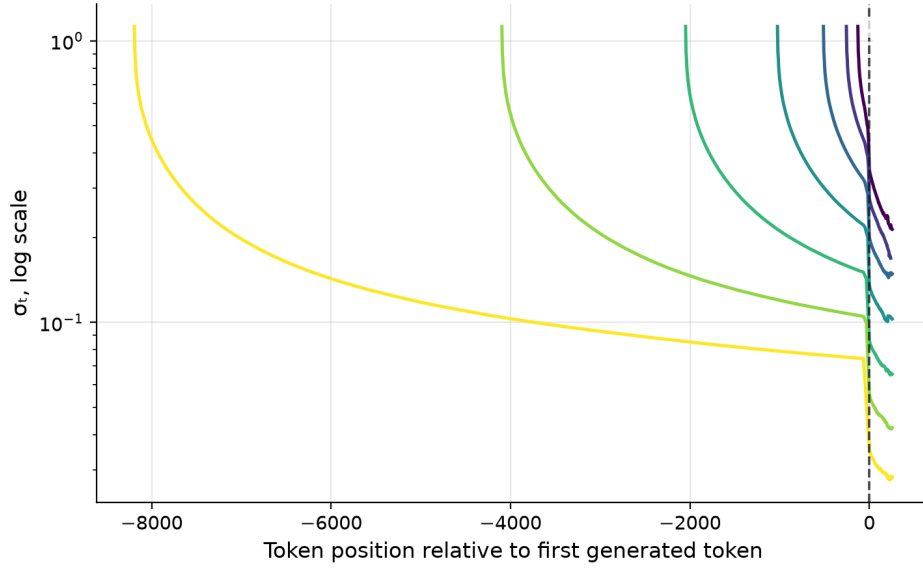
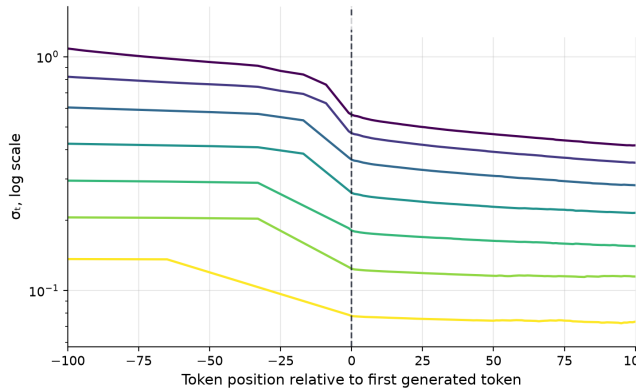
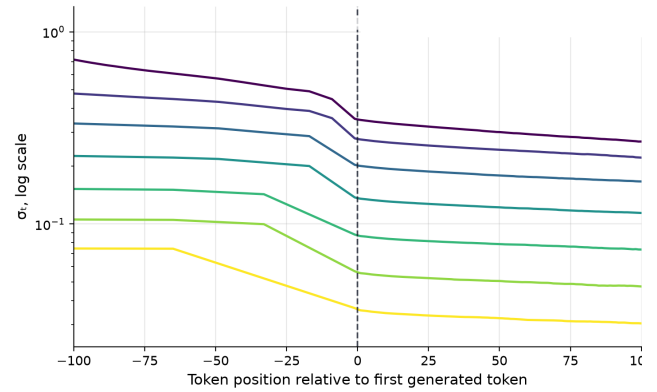


Figure 10. Additional effective-magnitude result for optimism/pessimism. As in Figure 4, longer prefixes leave a smaller correction at the beginning of generation, and a further drop occurs at the generation boundary.

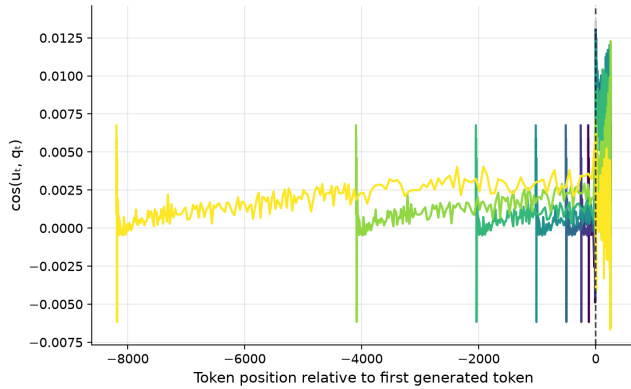


(a) Russian/English.

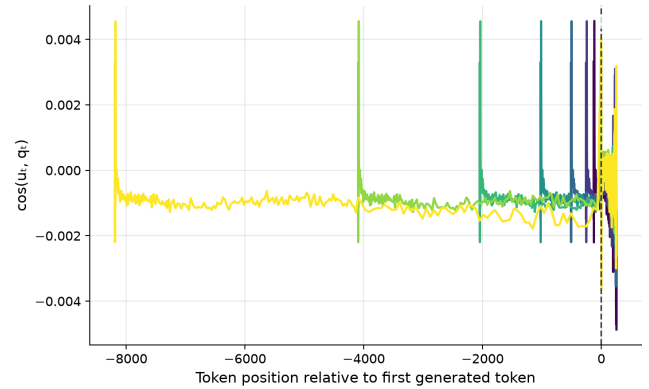


(b) Optimism/pessimism.

Figure 11. Leading singular magnitude near the prefill-to-generation boundary. Colors encode prefix length from 128 (purple) to 8192 (yellow).



(a) Russian/English, full trajectory.



(b) Optimism/pessimism, full trajectory.

Figure 12. Cosine alignment across prefill and generation. Colors encode prefix length from 128 (purple) to 8192 (yellow).

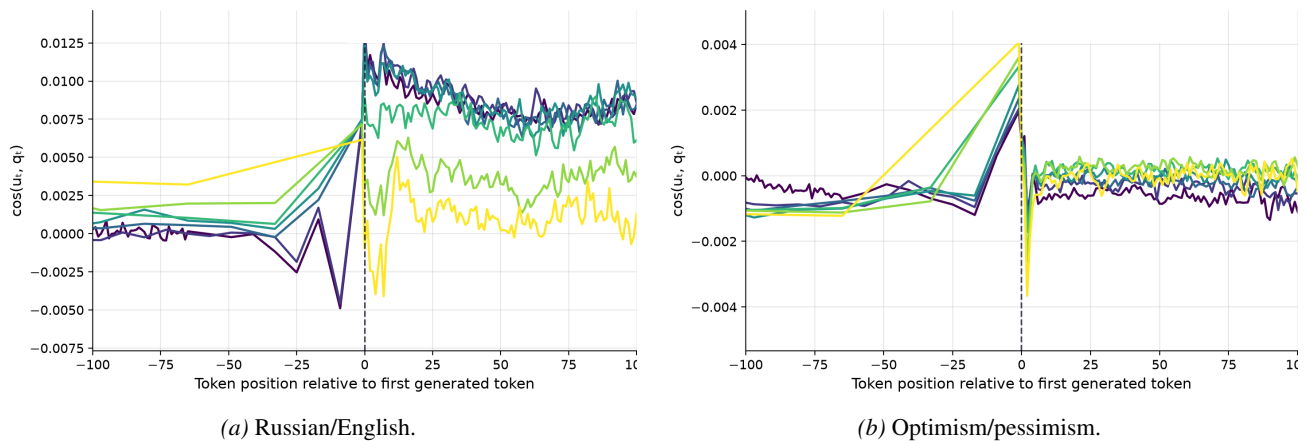


Figure 13. Cosine alignment near the generation boundary. The two concepts exhibit qualitatively different alignment even though both corrections contract in spectral magnitude.